Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

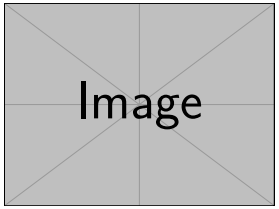
PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[english]babel
[]beamerthemeLuebeck helvet
listings java backgroundcolor=
, keywordstyle=, stringstyle=, identifierstyle=,
numberstyle=, language=Java cpp backgroundcolor=
, keywordstyle=, stringstyle=,
identifierstyle=, numberstyle=, language=C++ py backgroundcolor=
, keywordstyle=,
stringstyle=, language=Python js backgroundcolor=
, keywordstyle=, stringstyle=,
identifierstyle=, numberstyle=, language=JavaScript, escapechar=@ sh
basicstyle=, breaklines=true, commentstyle=, keywordstyle=,
stringstyle=, identifierstyle=, numberstyle=, language=bash
algorithm algpseudocode
xcolor



Beamer Template

Assets for LaTeX

Insitute

Author: Name Surname

2025/2026

Typography

- **Bold**
- *Italic*
- `Monospaced`
- Underlined
- Link

Lists

Itemize:

- **Item 1:** Example of an item in an itemize list.
- **Item 2:** Another item demonstrating the use of itemize.
- **Item 3:** Yet another example to show list formatting.

Lists

Enumerate:

1. **Item 1:** Example of an enumerated item.
2. **Item 2:** Another enumerated item to illustrate the format.
3. **Item 3:** Final item in the enumerated list.

Blocks

Note

This is a note block that provides additional information or comments about the topic discussed.

Example

Here is an example block that demonstrates how to use this particular format for presenting examples.

Alert

This alert block highlights important warnings or critical information that needs attention.

Mathematics

Let f and g be functions where $g : A \to B$ and $f : B \to \mathbb{R}$. Suppose g is differentiable at $x \in A$ and f is differentiable at $g(x)$. Then the composition function $f \circ g$ is differentiable at x, and its derivative is given by:

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x). \tag{1}$$

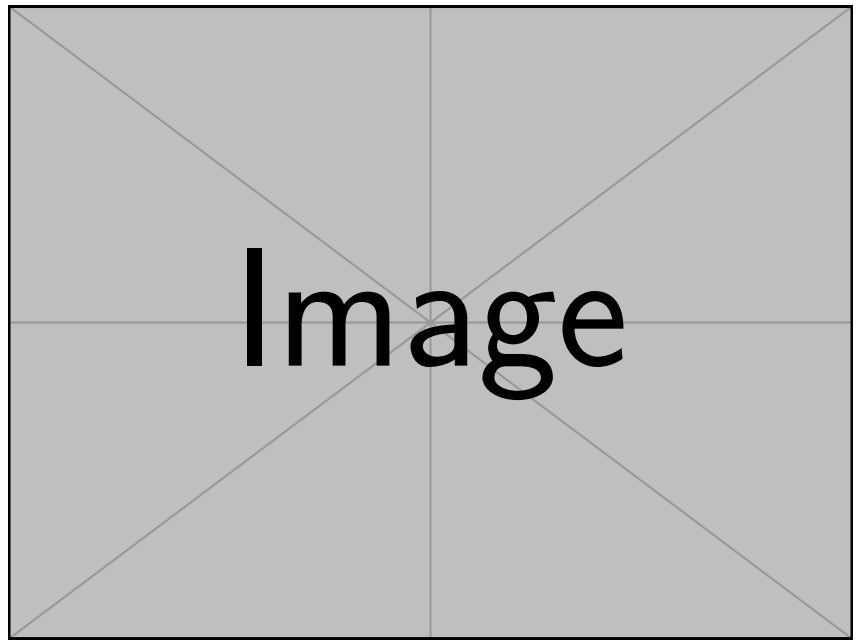
Code

style=py Model definition model = Sequential()
model.add(Flatten(input$_s$*hape* =
(28, 28)))*model*.*add*(*Dense*(128, *activation* =′
relu′))*model*.*add*(*Dense*(64, *activation* =′
relu′))*model*.*add*(*Dense*(10, *activation* =′
softmax′))*model*.*compile*(*optimizer* =′ *adam*′, *loss* =′
categorical$_c$*rossentropy*′, *metrics* = [′*accuracy*′])
Model Training history = model.fit(X$_t$*rain*, y_t*rain*, *epochs* =
10, *batch*$_s$*ize* = 32, *validation*$_s$*plit* = 0.2)
Model Evaluation test$_l$*oss*, *test*$_a$*cc* = *model*.*evaluate*(X_t*est*, y_t*est*)

Algorithms

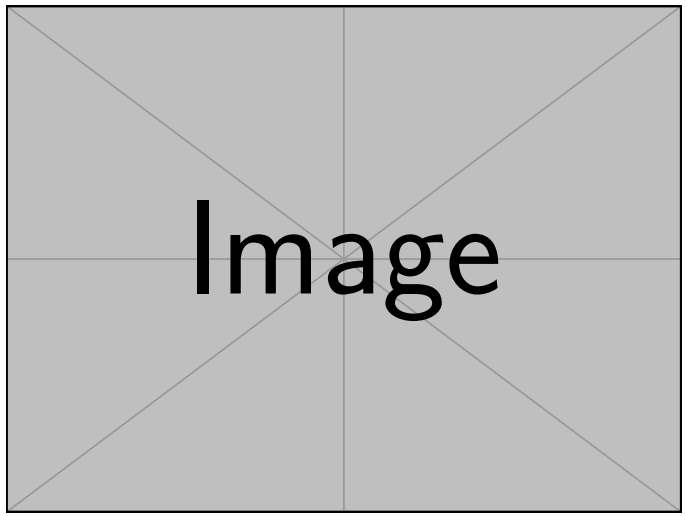
BubbleSort$array$ $n \leftarrow$ **length**$(array)$ $i \leftarrow 0$ **to** $n-1$ $j \leftarrow 0$ **to** $n-i-2$ $array[j] > array[j+1]$ **swap**$(array[j], array[j+1])$

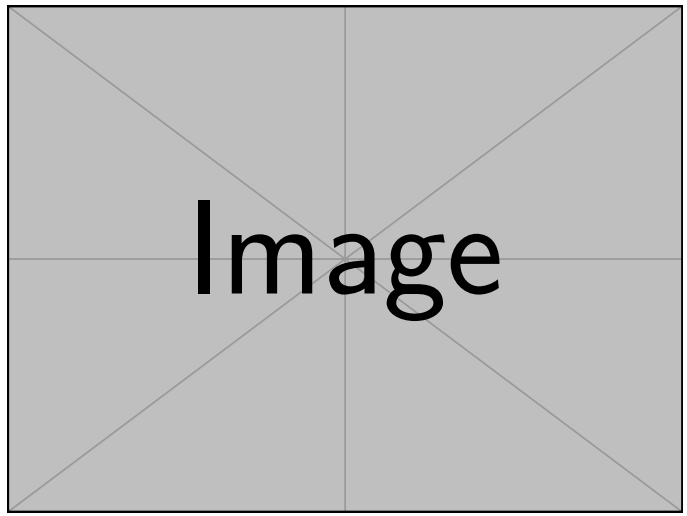
Images



Gambar: Example of single image.

Images





Tabel: Example of multiple images.

Tables

ID	Name	Surname	Birthdate
1	John	Doe	1990-05-15
2	Jane	Smith	1985-11-30
3	Alice	Johnson	2000-07-22
4	Bob	Brown	1992-02-09
5	Carol	Davis	1988-09-19

Tabel: Example of table.

Bibliography

Here is a reference to a source. For example, *Momentum Contrast for Unsupervised Visual Representation Learning by He et al. (2020)* [?] is a placeholder text for a bibliography entry.